

***Unrolled List –
Search Element
–Program Analysis***

Program

```
// Function to search for an element
void searchElement(int val) {
    if (head == NULL) {
        cout << "The list is empty.\n";
        return;
    }

    struct Node* temp = head;
    int nodeCount = 1;

    while (temp != NULL) {
        for (int i = 0; i < temp->numElements; i++) {
            if (temp->elements[i] == val) {
                cout << "Element " << val << " found in Node " <<
nodeCount << " at index " << i << ".\n";
                return;
            }
        }
        temp = temp->next;
        nodeCount++;
    }

    cout << "Element " << val << " not found in the list.\n";
}

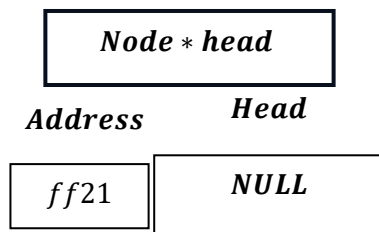
....

    case 4:
        cout << "Enter integer value to search: ";
        cin >> value;
        searchElement(value);
        break;
```

Program Analysis

1. List is Empty

```
if (head == NULL) {  
    cout << "The list is empty.\n";  
    return;  
}
```

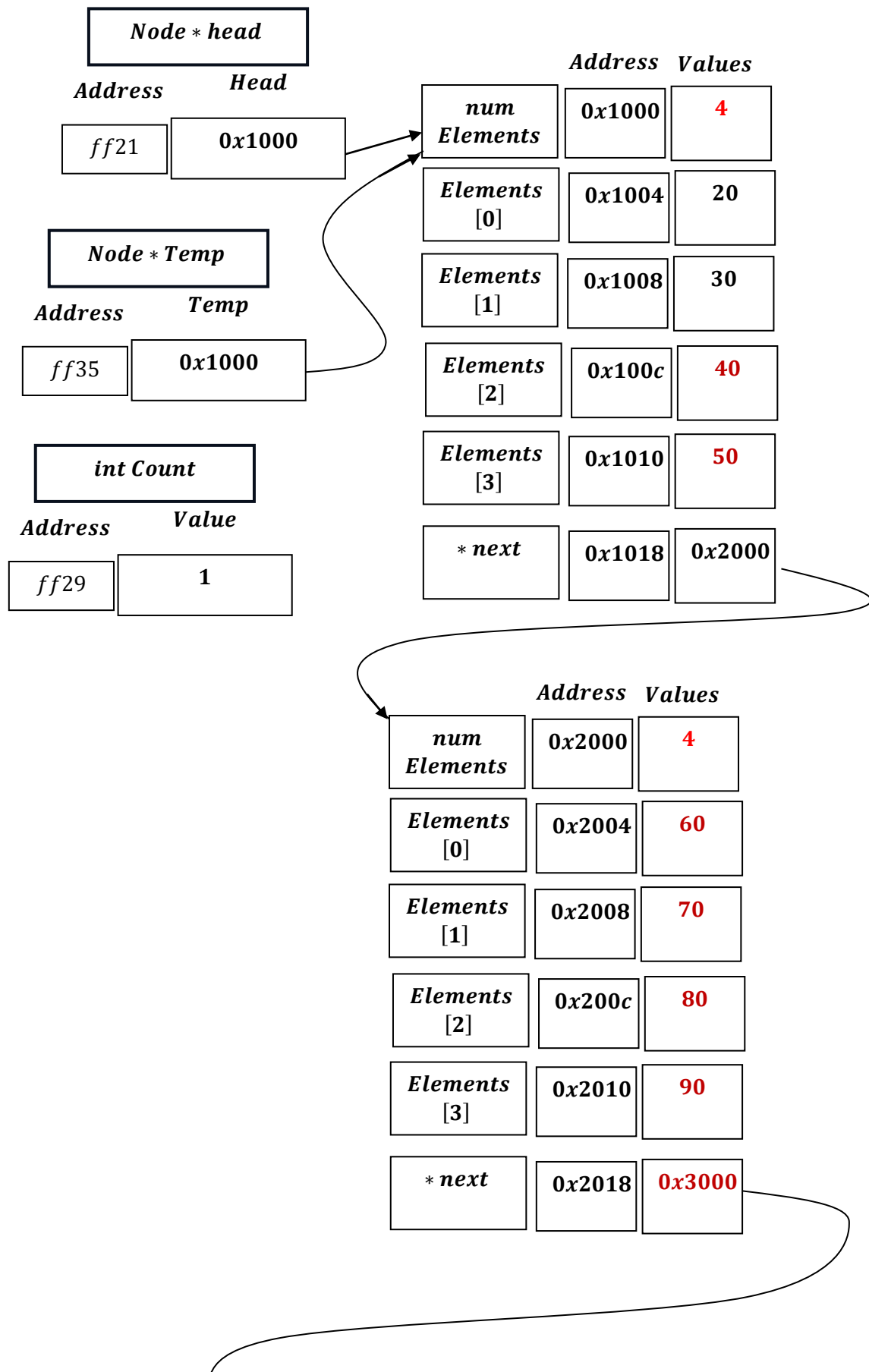


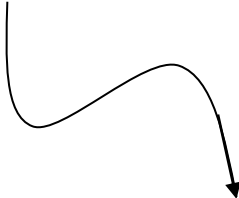
As head = NULL, it prints : `` The list is empty.``

Return void and exit from the function.

2. List is not Empty

```
struct Node* temp = head;  
int nodeCount = 1;
```





	<i>Address</i>	<i>Values</i>
<i>num Elements</i>	0x3000	4
<i>Elements</i> [0]	0x3004	100
<i>Elements</i> [1]	0x3008	110
<i>Elements</i> [2]	0x300C	120
<i>Elements</i> [3]	0x3010	130
<i>* next</i>	0x3018	NULL

```
cin >> value;
searchElement(value);
break;
```

Say, we are searching Element value : 60.

i. e. Function call: searchElement(value: 60).

```
while (temp != NULL) {
    for (int i = 0; i < temp->numElements; i++) {
        if (temp->elements[i] == val) {
            cout << "Element " << val << " found in Node " <<
nodeCount << " at index " << i << ".\n";
            return;
        }
    }
    temp = temp->next;
    nodeCount++;
}
```

while(*Temp*: 0x1000 ≠ *NULL*) $\left[\begin{array}{c} \textit{Condition} \\ \textit{is} \\ \textit{True} \end{array} \right] :$

for: *i* = 0; *i* < *temp* → *numElements*: 4 $\left[\begin{array}{c} \textit{Condition} \\ \textit{is} \\ \textit{True} \end{array} \right]$

if(*temp* → *elements*[0]: 20 = *val*: 60) $\left[\begin{array}{c} \textit{Condition} \\ \textit{is} \\ \textit{False} \\ \textit{and if block exits} \end{array} \right]$

i = *i* + 1 = 0 + 1 = 1.

for: *i* = 1; *i* < *temp* → *numElements*: 4 $\left[\begin{array}{c} \textit{Condition} \\ \textit{is} \\ \textit{True} \end{array} \right]$

if(*temp* → *elements*[1]: 30 = *val*: 60) $\left[\begin{array}{c} \textit{Condition} \\ \textit{is} \\ \textit{False} \\ \textit{and if block exits} \end{array} \right]$

i = *i* + 1 = 1 + 1 = 2.

for: *i* = 2; *i* < *temp* → *numElements*: 4 $\left[\begin{array}{c} \textit{Condition} \\ \textit{is} \\ \textit{True} \end{array} \right]$

if(*temp* → *elements*[2]: 40 = *val*: 60) $\left[\begin{array}{c} \textit{Condition} \\ \textit{is} \\ \textit{False} \\ \textit{and if block exits} \end{array} \right]$

i = *i* + 1 = 2 + 1 = 3.

$for: i = 3; i < temp \rightarrow numElements: 4 \left[\begin{array}{c} \text{Condition} \\ \text{is} \\ \text{True} \end{array} \right]$
 $if(temp \rightarrow elements[3]: 50 = val: 60) \left[\begin{array}{c} \text{Condition} \\ \text{is} \\ \text{False} \\ \text{and if block exits} \end{array} \right]$

$i = i + 1 = 3 + 1 = 4.$

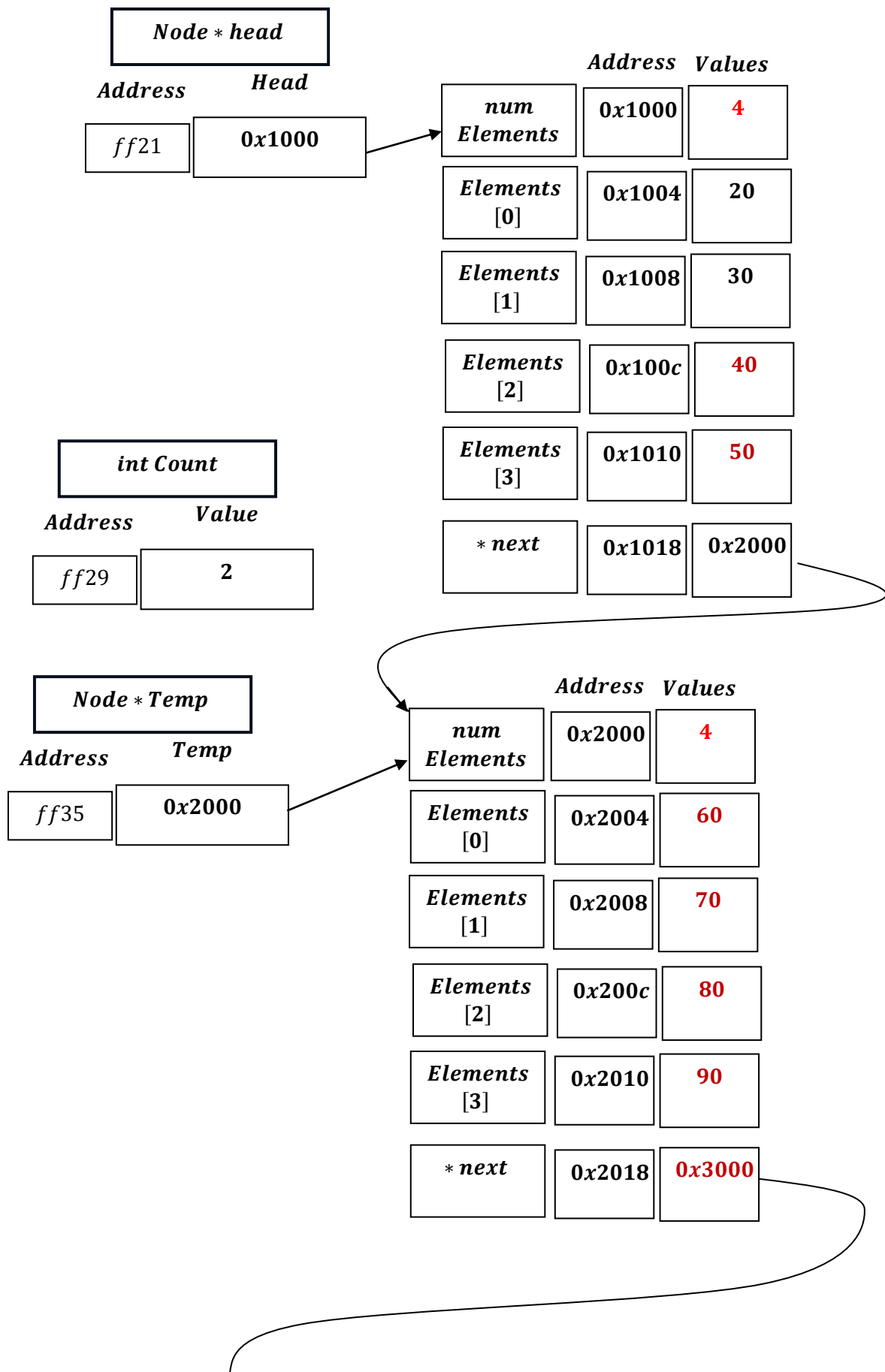
$for: i = 4; i < temp \rightarrow numElements: 4 \left[\begin{array}{c} \text{Condition} \\ \text{is} \\ \text{False} \\ \text{and for loop} \\ \text{Exits} \end{array} \right]$

Now,

```
temp = temp->next;
nodeCount++;
```

$Temp = Temp \rightarrow next = 0x2000$

$nodeCount = nodeCount + 1 = 1 + 1 = 2.$



	Address	Values
<i>num Elements</i>	0x3000	4
<i>Elements [0]</i>	0x3004	100
<i>Elements [1]</i>	0x3008	110
<i>Elements [2]</i>	0x300C	120
<i>Elements [3]</i>	0x3010	130
<i>* next</i>	0x3018	NULL

while(Temp: 0x2000 $\neq$ NULL) $\left[\begin{array}{c} \text{Condition} \\ \text{is} \\ \text{True} \end{array} \right]$:

for: i = 0; i < temp $\rightarrow$ numElements: 4 $\left[\begin{array}{c} \text{Condition} \\ \text{is} \\ \text{True} \end{array} \right]$

if(temp $\rightarrow$ elements[0]: 60 = val: 60) $\left[\begin{array}{c} \text{Condition} \\ \text{is} \\ \text{True} \\ \text{and if block runs} \end{array} \right]$

```
cout << "Element " << val << " found in Node " <<
nodeCount << " at index " << i << ".\n";
```

Print `` Element`` val: 60 ``found in Node`` nodeCount: 2 ``at index`` i: 0 .

```
return;
```

return will return void $\left[\begin{array}{c} \text{Return} \\ \text{Type} \end{array} \right]$ and exit from the function.

∴ Finally we have:

Element 60 found in Node 2 at index 0 .
